

```
1 students = {}
2
3
4 def add_student():
5     print("\n===== ADD STUDENT =====")
6
7     try:
8         roll_no = int(input("Enter Roll Number: "))
9     except ValueError:
10        print("Please enter a valid roll number.")
11        return
12
13    if roll_no in students:
14        print("Student already exists!")
15        return
16
17    name = input("Enter Student Name: ")
18    branch = input("Enter Branch: ")
19
20    try:
21        age = int(input("Enter Age: "))
22        python_marks = float(input("Enter Python Marks: "))
23        maths_marks = float(input("Enter Mathematics Marks: "))
24        english_marks = float(input("Enter English Marks: "))
25    except ValueError:
26        print("Please enter valid numbers.")
27        return
28
29    students[roll_no] = {
30        "name": name,
31        "branch": branch,
32        "age": age,
33        "python": python_marks,
34        "maths": maths_marks,
35        "english": english_marks
36    }
37
38    print("Student added successfully!")
39
40
41 def view_students():
42     print("\n===== ALL STUDENTS =====")
43
44     if not students:
45         print("No student records available.")
46         return
47
48     for roll_no, details in students.items():
49         print("\nRoll Number :", roll_no)
50         print("Name      :", details["name"])
51         print("Branch    :", details["branch"])
52         print("Age       :", details["age"])
53         print("Python    :", details["python"])
54         print("Mathematics :", details["maths"])
55         print("English   :", details["english"])
56         print("-----")
57
58
59 def search_student():
60     print("\n===== SEARCH STUDENT =====")
61
62     try:
63         roll_no = int(input("Enter Roll Number: "))
64     except ValueError:
65         print("Please enter a valid roll number.")
66         return
67
68     if roll_no in students:
69         details = students[roll_no]
70
71         print("\nStudent Found!")
72         print("Roll Number :", roll_no)
73         print("Name      :", details["name"])
74         print("Branch    :", details["branch"])
75         print("Age       :", details["age"])
```

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70
71     print("\nStudent Found!")
72     print("Roll Number :", roll_no)
73     print("Name          :", details["name"])
74     print("Branch         :", details["branch"])
75     print("Age            :", details["age"])
76     print("Python         :", details["python"])
77     print("Mathematics    :", details["maths"])
78     print("English        :", details["english"])
79 else:
80     print("Student not found.")
81
82
83 def update_student():
84     print("\n===== UPDATE STUDENT =====")
85
86     try:
87         roll_no = int(input("Enter Roll Number: "))
88     except ValueError:
89         print("Please enter a valid roll number.")
90         return
91
92     if roll_no not in students:
93         print("Student not found.")
94         return
95
96     name = input("Enter New Name: ")
97     branch = input("Enter New Branch: ")
98
99     try:
100         age = int(input("Enter New Age: "))
101         python_marks = float(input("Enter New Python Marks: "))
102         maths_marks = float(input("Enter New Mathematics Marks: "))
103         english_marks = float(input("Enter New English Marks: "))
104     except ValueError:
105         print("Please enter valid numbers.")
106         return
107
108     students[roll_no]["name"] = name
109     students[roll_no]["branch"] = branch
110     students[roll_no]["age"] = age
111     students[roll_no]["python"] = python_marks
112     students[roll_no]["maths"] = maths_marks
113     students[roll_no]["english"] = english_marks
114
115     print("Student details updated successfully!")
116
117
118 def delete_student():
119     print("\n===== DELETE STUDENT =====")
120
121     try:
122         roll_no = int(input("Enter Roll Number: "))
123     except ValueError:
124         print("Please enter a valid roll number.")
125         return
126
127     if roll_no in students:
128         del students[roll_no]
129         print("Student deleted successfully!")
130     else:
131         print("Student not found.")
132
133
134 def calculate_result():
135     print("\n===== STUDENT RESULT =====")
136
137     try:
138         roll_no = int(input("Enter Roll Number: "))
139     except ValueError:
140         print("Please enter a valid roll number.")
141         return
142

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142
143- if roll_no not in students:
144-     print("Student not found.")
145-     return
146
147 details = students[roll_no]
148
149 total = (
150     details["python"]
151     + details["maths"]
152     + details["english"]
153 )
154
155 percentage = total / 3
156
157 print("\nStudent Name :", details["name"])
158 print("Roll Number   :", roll_no)
159 print("Total Marks   :", total)
160 print("Percentage    :", round(percentage, 2), "%")
161
162- if percentage >= 90:
163-     grade = "A+"
164- elif percentage >= 80:
165-     grade = "A"
166- elif percentage >= 70:
167-     grade = "B"
168- elif percentage >= 60:
169-     grade = "C"
170- elif percentage >= 50:
171-     grade = "D"
172- else:
173-     grade = "F"
174
175 print("Grade         :", grade)
176
177- if percentage >= 40:
178-     print("Result      : PASS")
179- else:
180-     print("Result      : FAIL")
181
182
183- def find_topper():
184-     print("\n===== TOPPER =====")
185
186-     if not students:
187-         print("No student records available.")
188-         return
189
190     topper_roll = None
191     highest_percentage = -1
192
193-     for roll_no, details in students.items():
194-         total = (
195-             details["python"]
196-             + details["maths"]
197-             + details["english"]
198-         )
199
200         percentage = total / 3
201
202-         if percentage > highest_percentage:
203-             highest_percentage = percentage
204-             topper_roll = roll_no
205
206     topper = students[topper_roll]
207

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206     topper = students[topper_roll]
207
208     print("\nTopper Details")
209     print("-----")
210     print("Roll Number :", topper_roll)
211     print("Name       :", topper["name"])
212     print("Branch      :", topper["branch"])
213     print("Percentage  :", round(highest_percentage, 2), "%")
214     print("-----")
215
216
217 def count_students():
218     print("\n===== TOTAL STUDENTS =====")
219     print("Total Students:", len(students))
220
221
222 def main():
223     while True:
224         print("\n=====")
225         print("          STUDENT MANAGEMENT SYSTEM")
226         print("=====")
227         print("1. Add Student")
228         print("2. View All Students")
229         print("3. Search Student")
230         print("4. Update Student")
231         print("5. Delete Student")
232         print("6. Calculate Result")
233         print("7. Find Topper")
234         print("8. Count Students")
235         print("9. Exit")
236         print("=====")
237
238         choice = input("Enter your choice: ")
239
240         if choice == "1":
241             add_student()
242
243         elif choice == "2":
244             view_students()
245
246         elif choice == "3":
247             search_student()
248
249         elif choice == "4":
250             update_student()
251
252         elif choice == "5":
253             delete_student()
254
255         elif choice == "6":
256             calculate_result()
257
258         elif choice == "7":
259             find_topper()
260
261         elif choice == "8":
262             count_students()
263
264         elif choice == "9":
265             print("\nThank you for using Student Management System!")
266             print("Goodbye!")
267             break
268
269         else:
270             print("Invalid choice! Please enter 1 to 9.")
271
272
273 main()

```

```
=====
                STUDENT MANAGEMENT SYSTEM
=====
```

1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Calculate Result
7. Find Topper
8. Count Students
9. Exit

```
=====
Enter your choice: 1
```

```
===== ADD STUDENT =====
```

Enter Roll Number: 265626078

Enter Student Name: Nandani Tomar

Enter Branch: cse cloud computing and au

Enter Age: 18

Enter Python Marks: 98

Enter Mathematics Marks: 97

Enter English Marks: 99

Student added successfully!

```
Enter Branch: cse cloud computing and au
Enter Age: 18
Enter Python Marks: 98
Enter Mathematics Marks: 97
Enter English Marks: 99
Student added successfully!
```

```
=====
                        STUDENT MANAGEMENT SYSTEM
=====
```

- ```
=====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Calculate Result
7. Find Topper
8. Count Students
9. Exit
=====
```

```
Enter your choice: 9
```

```
Thank you for using Student Management S
Goodbye!
```